

FUNCTIONAL REQUIREMENTS

R1: Question Bank and Adversarial Repository Management (UC1)

R1.1: Question Creation

R1.1.1: Allow a recruiter to create a new source question.

R1.1.2: Allow a recruiter to categorise a question by category, difficulty level, question type, and tags during creation.

R1.1.3: Allow a recruiter to store all created source questions in the question bank.

R1.2: Question Editing

R1.2.1: Allow a recruiter to edit the title, content, category, difficulty level, question type, and tags of an existing source question.

R1.3: Question Deletion

R1.3.1: Allow a recruiter to delete a source question from the question bank.

R1.4: Question Search and Filtering

R1.4.1: Allow a recruiter to search and filter the question bank by keyword, category, difficulty level, question type and tags.

R1.5: Repository Management

R1.5.1: Maintain a source question repository using the Question Bank table to store all original questions.

R1.5.2: Maintain a separate adversarial question repository using the Adversarial Questions table to store all generated adversarial versions.

R1.5.3: Maintain a reference between the source question and adversarial question.

R2: Adversarial Question Generation (UC2)

R2.1: Generation Triggering

R2.1.1: Allow a recruiter to trigger adversarial question generation for one or more selected source questions within an assessment.

R2.2: Question Generation Process

R2.2.1: Retrieve the selected source question content from the question bank table when adversarial generation is triggered.

R2.2.2 Construct a structured adversarial prompt template and submit it along with the source question content to the Google Gemini API.

R2.2.3 Create an adversarial question using LLM which will embed adversarial characteristics into the question.

R2.2.4 Display the adversarial version of a question to the candidate during assessment delivery.

R2.3: Question Generation Preview and Review

R2.3.1: Allow the recruiter to regenerate an adversarial version if the

initial result is unsatisfactory.

R3: AI Usage Similarity Detection (UC3)

R3.1: Detection Triggering

R3.1.1: Automatically trigger the AI usage detection pipeline upon submission of a completed assessment.

R3.2: Data Retrieval

R3.2.1: Retrieve the corresponding adversarial question and submitted answer for each candidate response.

R3.3: Reference Answer Generation

R3.3.1: Submit the adversarial question to the Google Gemini API and retrieve a reference answer representing what the LLM would produce for that question.

R3.4: Similarity Computation

R3.4.1: Compute a similarity score measuring the similarity in phrasing patterns, structure, and formatting between the candidate's answer and the LLM reference answer.

R3.5: Results Storage

R3.5.1: Store full full analysis result in the AI analysis table.

R3.5.2: Ensure the ability to support future extensibility to multiple LLM providers.

R3.6: Results Display

R3.6.1: Display the AI usage probability score on the recruiter's results dashboard.

R3.6.2: The system shall display a similarity score breakdown showing semantic and structural similarity scores individually.

R4: Improved Complete Assessment Experience (UC4)

R4.1: Assessment Visibility

R4.1.1: Allow a candidate to view all assessments that have been assigned to them on the candidate dashboard

R4.2: Assessment Resumption

R4.2.1: Allow a candidate to resume an incomplete assessment from the point at which they left off.

R4.3: Question Answering

R4.3.1: Allow a candidate to answer coding questions through an integrated code editor supporting Python.

R4.4: Code execution

R4.4.1: Upon submission of a coding question, the system should

send the candidate's Python code to Judge0 CE for execution.

R4.4.2 Retrieve and execute appropriate test cases against the candidate's submission.

R4.5: Scoring Processing and Storage

R4.5.1: Calculate and update scoring information for individual responses and overall assessments.

R4.6: Assessment Submission

R4.6.1: Allow a candidate to submit their completed assessment.

R4.6.2: Trigger the AI usage detection pipeline automatically upon assessment submission.

R5: Custom Assessment Creation and Assignment (UC5)

R5.1: Assessment Creation

R5.1.1: Allow a recruiter to create a new assessment by providing a title, description and duration.

R5.2: Question Selection

R5.2.1: Allow a recruiter to filter the question bank by category, difficulty level, and question type when selecting questions for an assessment.

R5.2.2: Allow a recruiter to select one or more source questions from the question bank to include in the assessment.

R5.3: Adversarial Generation Trigger

R5.3.1: Allow a recruiter to trigger adversarial question generation for all selected questions from within the assessment builder.

R5.4: Assessment Publishing

R5.4.1: Allow the recruiter to publish and finalise assessment creation.

R5.5: Assessment Assignment

R5.5.1: Allow a recruiter to assign a completed assessment to one or more candidates.

R5.5.2: Generate a unique access token per candidate assessment assignment.